

Problem 6

Problem 1 Sketch a graph of the function f defined by

$$f(x) = \begin{cases} 2x - 2 \tan^{-1}(x) - x \ln(x^2 + 1) & \text{for } x \leq 1 \\ \sqrt{x-2} & \text{for } x \geq 2 \end{cases}$$

- (a) Find the domain of f .

Explanation. The domain of f is $(-\infty, 1] \cup [2, \infty)$.

- (b) Show that $f'(x) = -\ln(x^2 + 1)$ on the interval $(-\infty, 1)$.

Explanation. For $x < 1$ we have

$$\begin{aligned} f'(x) &= 2 - \frac{2}{1+x^2} - \ln(x^2 + 1) - x \cdot \frac{1}{x^2 + 1} \cdot 2x \\ &= 2 - \frac{2}{1+x^2} - \ln(x^2 + 1) - \frac{2x^2}{x^2 + 1} \\ &= 2 - \frac{2+2x^2}{x^2 + 1} - \ln(x^2 + 1) \\ &= 2 - 2 \cdot \frac{1+x^2}{x^2 + 1} - \ln(x^2 + 1) \\ &= 2 - 2 - \ln(x^2 + 1) = -\ln(x^2 + 1) \end{aligned}$$

- (c) Compute a formula for f' on the interval $(2, \infty)$.

Explanation. For $x > 2$ we have

$$f'(x) = \frac{1}{2\sqrt{x-2}}$$

- (d) Compute a formula for f'' on the interval $(-\infty, 1)$.

Explanation. For $x < 1$ we have

$$f''(x) = \frac{-1}{x^2 + 1} \cdot 2x = \frac{-2x}{x^2 + 1}$$

- (e) Compute a formula for f'' on the interval $(2, \infty)$.

Explanation. For $x > 2$ we have

$$f''(x) = \frac{-1}{4(x-2)^{3/2}}$$

- (f) List all the asymptotes (with limit justifications, if appropriate). (If f does not have a particular asymptote write "NONE".)

(i) Vertical asymptotes:

Explanation. NONE: For $x \geq 2$, $f(x) = \sqrt{x-2}$ so no vertical asymptotes there. For $x \leq 1$, $f(x) = 2x - 2 \tan^{-1}(x) - x \ln(x^2 + 1)$ and we can look at each term. The term $2x$ is a polynomial. The term $-2 \tan^{-1}(x)$ is bounded. Finally, the term $-x \ln(x^2 + 1)$. We would be concerned about where $x^2 + 1 = 0$ but that never happens because of the squared term.

(ii) Horizontal asymptotes:

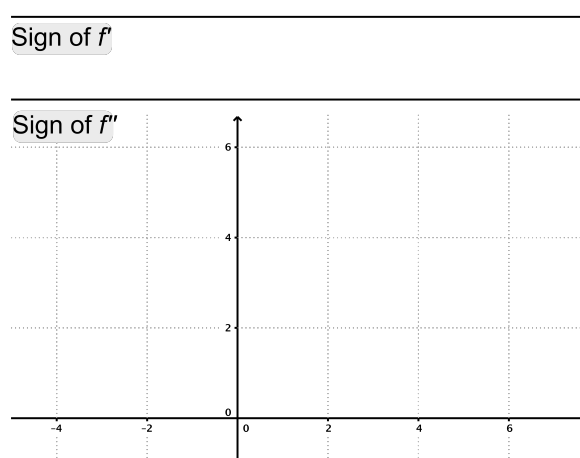
Explanation. No H.A. as $x \rightarrow \infty$:

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \underbrace{\sqrt{x-2}}_{\text{form } \infty} = \infty$$

No H.A. as $x \rightarrow -\infty$:

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} (2x - 2 \tan^{-1}(x) - x \ln(x^2 + 1)) \\ &= \lim_{x \rightarrow -\infty} [x \cdot (2 - \ln(x^2 + 1)) - 2 \tan^{-1}(x)] \\ &= \lim_{x \rightarrow -\infty} [x \cdot (2 + \ln(x^2 + 1)^{-1}) - 2 \tan^{-1}(x)] \\ &= \lim_{x \rightarrow -\infty} \left[\underbrace{x}_{\rightarrow -\infty} \cdot \left(2 + \ln \left(\underbrace{\frac{1}{x^2 + 1}}_{\rightarrow 0} \right) \right) - \underbrace{2 \tan^{-1}(x)}_{\rightarrow -\pi/2} \right] \\ &= \lim_{x \rightarrow -\infty} \left[\underbrace{x}_{\rightarrow -\infty} \cdot \left(2 + \underbrace{\ln \left(\frac{1}{x^2 + 1} \right)}_{\rightarrow -\infty} \right) - \underbrace{2 \tan^{-1}(x)}_{\rightarrow -\pi} \right] \\ &= \infty \end{aligned}$$

- (g) In the following sign chart for f' , list the points (in the domain of f) where $f'(x) = 0$ or $f'(x)$ is undefined, indicate the sign of f' on each subinterval, and identify any local extrema (with "local max" or "local min".) In the following sign chart for f'' , list the points (in the domain of f) where $f''(x) = 0$ or $f''(x)$ is undefined, indicate the sign of f'' on each subinterval, and identify any inflection points. Use this information to sketch a graph of f in the blank grid. (Be sure to indicate any x - or y -intercepts.)



Explanation. Finding critical points of f :

For $x < 1$ we have

$$\begin{aligned} f'(x) = 0 &\iff -\ln(x^2 + 1) = 0 \\ &\iff \ln(x^2 + 1) = 0 \\ &\iff x^2 + 1 = e^0 = 1 \\ &\iff x = 0. \end{aligned}$$

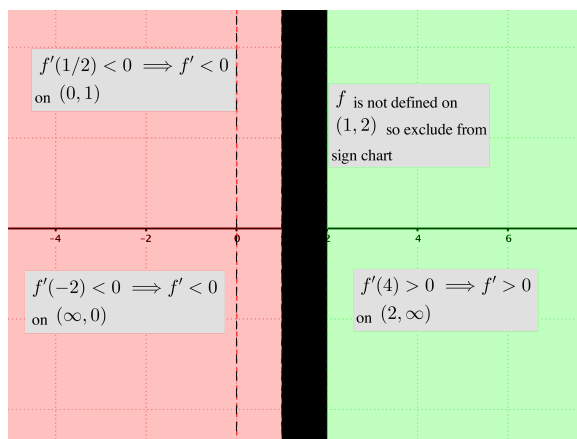
Critical point at 0 when $x < 1$.

For $x > 2$ we have

$$f'(x) = 0 \iff \frac{1}{2\sqrt{x-2}} = 0$$

This last equation is never true. Hence no critical points when $x > 2$.

Sign chart of f' :



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Using the sign chart above, we see that the sign of f' does not change at $x = 0$ and is therefore neither a maximum or minimum. So f does not have a local maximum or local minimum.

Using the sign chart above, we see f' is positive on $(2, \infty)$ and thus intervals where f is increasing: $(2, \infty)$

Using the sign chart above, we see f' is negative on $(-\infty, 0)$, $(0, 1)$ and thus intervals where f is decreasing: $(-\infty, 0)$, $(0, 1)$.

Find candidates for inflection points:

For $x < 1$ we have

$$\begin{aligned} f''(x) = 0 &\iff \frac{-2x}{x^2 + 1} = 0 \\ &\iff x = 0. \end{aligned}$$

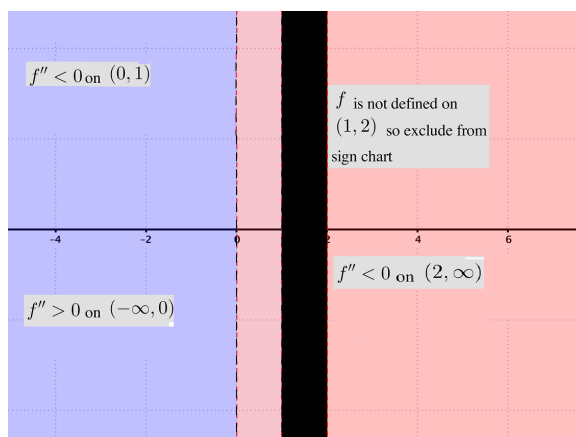
So, $x = 0$ is a candidate for an inflection point.

For $x > 2$ we have

$$f''(x) = 0 \iff \frac{-1}{4(x-2)^{3/2}} = 0$$

This last equation has no solution. Hence no candidates for inflection points when $x > 2$.

We need to determine the sign of f'' on the intervals $(-\infty, 0)$, $(0, 1)$, $(2, \infty)$.



f'' changes sign at $x = 0$ so $(0, 0)$ is an inflection point.

intervals where f is concave up: $(-\infty, 0)$

intervals where f is concave down: $(0, 1)$, $(2, \infty)$

Final sketch of graph:

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